The f0(x, lambda-prime) connection formula for the generalized Zwegers mu-function

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doi:10.3842/SIGMA.2023.014; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2023-014-a-generalization-of-zwegers-mu-function-acco/pinned-original-source.tex (A Generalization of Zwegers' mu-Function According to the q-Hermite-Weber Difference Equation); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author substitutes lambda -> lambda-prime in the connection formula amuconnect (2.2), eliminates g0(x) from the resulting two-by-two system eq:different variable (2.5), and closes the remaining theta-ratio identity by the pole-residue constancy argument for the auxiliary function C(x) (residue at x=1 vanishes, C(xq)=C(x), evaluate at x=-a/lambda). In-span labels render from the card's own counters: (2.4) eq:tuchimi connection (statement equation, label line 790; the first row of its gather is nonumber and renders unnumbered exactly as in the source) and (2.5) eq:different variable (proof equation, line 799). Out-of-span navigation pointer declared as external reference: amuconnect (2.2, line 675, used at line 797). The proof tail carries the author's own \tag*{\qed} plus \renewcommand{\qed}{} device (source lines 820-821), copied verbatim. Seed convention: counters stepped from the source itself and cross-validated against a full host compile of the pinned source (aux ground truth): section 2 of the shared [thm] counter holds lem:lemma1 (2.1) and lem:lemma2 (2.2) before this theorem, and section 2 holds numbered equations (2.1) eq:(26), (2.2) amuconnect and (2.3) eq:prot connec before the span, so the card prints Theorem 2.3 with in-span equations (2.4) and (2.5).

Editorial note. The label thm:tuchimi connection names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the article source.

Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, no bibliography in the counted span (the span carries no \cite commands).

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- The connection formula amuconnect (2.2), source line 675, for f0(x, lambda) in terms of f-infinity(x) and g-infinity(x) with theta-q-function coefficients; the theorem's proof substitutes lambda -> lambda-prime into it.
- The Jacobi theta function theta_q and the q-Pochhammer symbols (a)-infinity, (q)-infinity appearing in the theta-coefficients; the proof uses only the pole positions of the auxiliary function C(x) and its periodicity C(xq)=C(x), both stated inside the span (source line 812).
- The elementary elliptic-function argument that a q-periodic meromorphic function whose prescribed-pole residues vanish is constant, with the constant then read off at one further point (source lines 812-821).

Source reference. Original equation 2.2: sources/sigma-2023-014-a-generalization-of-zwegers-mu-function-acco/pinned-original-source.tex, source lines 675–675. Navigation pointer invoked at line 797: the counted text cites that numbered equation of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it; the cited equation is an allowed input and none of it is reproduced inside the span.

Theorem 2.3. *The following equation holds:*

$$\begin{aligned} f_0(x,\lambda') &= \frac{\theta_q(\lambda')\theta_q(ax/\lambda')x^\alpha}{\theta_q(\lambda'/a)\theta_q(x/\lambda')a} f_\infty(x,\lambda) \\ &- (a)_\infty (q)_\infty^2 \frac{\theta_q(-\lambda'/x\lambda)\theta_q(-x)\theta_q(-\lambda\lambda'/a)}{\theta_q(x\lambda)\theta_q(\lambda'/x)\theta_q(\lambda'/a)\theta_q(a/\lambda)} g_\infty(x). \end{aligned} \tag{2.4}$$

Proof. We replace the variable λ by λ' in the connection formula (2.2) of the case of $f_0(x,\lambda)$:

$$\begin{pmatrix} f_0(x,\lambda') \\ f_\infty(x,\lambda) \end{pmatrix} = -\frac{(q)_\infty}{(q/a)_\infty} \begin{pmatrix} \dfrac{\theta_q(\lambda')\theta_q(ax/\lambda')x^\alpha}{\theta_q(\lambda'/a)\theta_q(x/\lambda')a} & 1 \\ 1 & \dfrac{\theta_q(\lambda)\theta_q(x\lambda/a)}{\theta_q(\lambda/a)\theta_q(x\lambda)}x^{-\alpha} \end{pmatrix} \begin{pmatrix} g_0(x) \\ g_\infty(x) \end{pmatrix}. \tag{2.5}$$

By erasing the function $g_0(x)$ in the above (2.5), we have

$$\begin{aligned} & f_0(x,\lambda') - \frac{\theta_q(\lambda')\theta_q(ax/\lambda')x^\alpha}{\theta_q(\lambda'/a)\theta_q(x/\lambda')a} f_\infty(x,\lambda) \\ &= \frac{(q)_\infty}{(q/a)_\infty} \left\{ \frac{\theta_q(\lambda)\theta_q(\lambda')\theta_q(x\lambda/a)\theta_q(ax/\lambda')}{a\theta_q(\lambda/a)\theta_q(\lambda'/a)\theta_q(x\lambda)\theta_q(x/\lambda')} - 1 \right\} g_\infty(x) \\ &= \frac{(a)_\infty (q)_\infty^2}{\theta_q(-a)} \frac{\theta_q(-\lambda'/x\lambda)\theta_q(-x)}{\theta_q(x\lambda)\theta_q(\lambda'/x)} C(x) g_\infty(x), \end{aligned}$$

where

$$C(x) = \frac{\lambda'}{ax} \frac{\theta_q(\lambda)\theta_q(\lambda')\theta_q(x\lambda/a)\theta_q(ax/\lambda')}{\theta_q(-\lambda'/x\lambda)\theta_q(-x)\theta_q(\lambda/a)\theta_q(\lambda'/a)} - \frac{\theta_q(x\lambda)\theta_q(\lambda'/x)}{\theta_q(-\lambda'/x\lambda)\theta_q(-x)}.$$

The function $C(x)$ has a simple pole in the points $x = q^j, \frac{\lambda'}{\lambda}q^j$ for some $j \in \mathbb{Z}$, and satisfies $C(xq) = C(x)$. We calculate the residue at the pole $x = 1$,

$$\begin{aligned}\lim_{x\to 1} \theta_q(-x)C(x) &= \frac{\lambda'}{a}\frac{\theta_q(\lambda)\theta_q(\lambda')\theta_q(\lambda/a)\theta_q(a/\lambda')}{\theta_q(-\lambda'/\lambda)\theta_q(\lambda/a)\theta_q(\lambda'/a)} - \frac{\theta_q(\lambda)\theta_q(\lambda')}{\theta_q(-\lambda'/\lambda)} \\ &= \frac{\theta_q(\lambda)\theta_q(\lambda')}{\theta_q(-\lambda'/\lambda)}\left\{\frac{\lambda'}{a}\frac{\theta_q(a/\lambda')}{\theta_q(\lambda'/a)} - 1\right\} = 0.\end{aligned}$$

It follows that the doubly periodic function $C(x)$ is constant in x. Hence, we have

$$C(x) = C(-a/\lambda) = -\frac{\theta_q(-a)\theta_q(-\lambda\lambda'/a)}{\theta_q(\lambda'/a)\theta_q(a/\lambda)}. \qquad \square$$